

CCIE Automation Lab Exam Revision 1.1

Blueprint Revisions

Today, products and technologies are evolving faster than ever before. To keep up with the fast pace, we are introducing a new agile process that will allow us to align our exams faster with these changes: *minor revisions*. Minor revisions will provide us with the agility and speed that are necessary to adjust our programs to match industry changes and the evolution of technologies. They will do so, by allowing us to update track details (exam blueprint, equipment list, and software) more frequently while keeping overall changes to a minimum (smaller than 20%). This not only allows us to ensure our content stays relevant but also, minimizes learning curves in between revisions.

The main objective of a minor revision is to:

- Further scope out the exam blueprint, by ensuring exam objectives are clear.
- Introduce new blueprint tasks to ensure exam relevancy as relevant today.
- Phase out old(er) products and/or technology solutions that are less relevant today.
- Update equipment and/or software.

Today we are revising the CCIE Automation Exam.

CCIE Automation Lab Exam, minor revision 1.1

The CCIE Automation lab exam will be updating through a *minor revision* (changes are small and incremental). Although the overall domains within the exam blueprint might look similar at first sight, with this minor revision we both added and removed technology solutions to ensure exam relevancy.

Please, refer to www.cisco.com/go/CertRoadmap for the list of exam topics covered in the updated CCIE Automation lab exam and for more information regards to the CCIE Automation certification program. Candidates can expect to be tested on the new exam material for CCIE Automation starting August 20, 2024.

CCIE Automation Lab Exam (v1.1) – Executive Summary

The new minor revision for CCIE Automation allows us to keep the domains and topics closely aligned with today's software development technologies and solutions. The CCIE Automation v1.1 lab exam blueprint introduces an increased focus on diagnosing application performance issues, using tools such as AppDynamics and Thousandeyes. Additionally interaction with new APIs received an update by adding GraphQL. To increase clarity for the CCIE Automation candidate, and to reflect current technological evolutions, some tasks have been removed from the exam topics.

Below is a comprehensive summary of the CCIE Automation v1.1 lab exam topics changes.

CCIE Automation Lab Exam Revision 1.1

Exam Topics Changes Highlights: (Please note these are highlights only therefore are not all-inclusive, refer to the actual Exam Topics list for details)

V1.0	V1.1
1. Software Design, Development, and Deployment (20%)	1. Software Design, Development, and Deployment (20%)
1.2 Recommend the deployment strategy to mitigate risk and impact on service availability, for a given scenario	1.5 Diagnose application performance issues - such as asynchronous request processing, database delays, high memory and CPU utilization, microservice network delays, and asymmetric routing - using network and application tools as well as assurance data.
2. Infrastructure as Code (30%)	2. Infrastructure as Code (30%)
2.1 Create a scalable solution for infrastructure automation (considering areas such as network impact, risk, and tool selection)	2.1 Build, manage, and operate a Python-based REST API with a web application framework (endpoints, HTTP request and response, OpenAPI specification)
	2.3 Consume and use a new API, given the documentation
	2.3.a REST
	2.3.b GraphQL
3. Network Programmability and Automation (25%)	3. Network Programmability and Automation (25%)
3.3 Deploy an application on a Cisco IOS XE device leveraging the technologies of Guest Shell and app-hosting	3.1 Create, modify, and troubleshoot scripts by using Python libraries and SDK documentation to automate against APIs (ACI, AppDynamics, DNA Center Catalyst Center, FDM, Intersight, IOS XE, Meraki, NSO, Webex)
4. Containers (10%)	4. Containers (10%)
5. Security (15%)	5. Security (15%)



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Hardware and Software Equipment

In support of the updated CCIE Automation revision 1.1 the lab exam environment, minor updates were made to both the equipment and software releases used. Candidates who want to prepare for the exam are now advised to use the following Cisco equipment and Software releases. Visit CCIE Automation Lab [Exam Equipment and Software v1.1](#) for a complete overview.

V1.0	V1.1
	Cloud
	• Cisco AppDynamics • Cisco ThousandEyes

Exam format

No changes have been made to the lab exam format in this minor revision. Visit CCIE Automation [Exam Topics](#) for more information about the lab exam format.

